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Homework 7

1.

Example7.4.png

Exercise 7.5 Given the normal target $\mathcal{N}_p(0, (1-\rho)I + \rho J)$:

a. Write a Gibbs sampler using the conditional distributions provided in Example 7.4. Run your `R` code for $p = 5$ and $\rho = .25$, and verify graphically that the marginals are all $\mathcal{N}(0, 1)$.
b. Compare your algorithm using $T = 500$ iterations with `rmnorm` described in Section 2.2.1 in terms of execution time.
c. Propose a constrained subset that is not a hypercube, and derive the corresponding Gibbs sampler. (*Hint*: Consider, for example, a constraint such as $\sum_{i=1}^{m} x_i^2 \leq \sum_{i=m+1}^{p} x_i^2$ for $m \leq p-1$.)

(a)

We are given the target distribution $X \sim N_p(0, (1-\rho)I + \rho J)$. From Example 7.4, the conditional distribution of X_i given all other components is said to be

$$X_i | x_{(-i)} \sim N(\frac{(p-1)\rho}{1+(p-2)\rho}\bar{x}_{(-i)}, \frac{1+(p-2)\rho-(p-1)\rho^2}{1+(p-2)\rho})$$

which we'll use in the code below, following the algorithm of drawing conditionally based on previously drawn values.

```
> # 1. (a)
> n = 500
> p = 5
> rho = 0.25
> gibbs_sample = function(n, p, rho) {
+   # Precomputed constants for conditional distribution from Example 7.4
+   conditional_mean_coeff = ((p - 1) * rho) / (1 + ((p - 2) * rho))
+   sigma_squared = (1 + ((p - 2) * rho) - ((p - 1) * rho^2)) / (1 + ((p
- 2) * rho))
+   sigma = sqrt(sigma_squared)
+
+   X <- array(0, dim = c(n, p)) # init arrays
+   X[1, ] <- rnorm(p) # init chains
+
+   for (t in 2:n) {
+     x = X[t - 1, ]
+     for (i in 1:p) {
+       x_mean_minus = (sum(x) - x[i]) / (p - 1)
+       x[i] = rnorm(1, mean = conditional_mean_coeff * x_mean_minus, sd
= sigma)
+     }
+     X[t, ] = x
+   }
```

```
+
+   X
+ }
> X = gibbs_sample(n, p, rho)
> hist(X[, p], breaks = 50, probability = TRUE,
+      main = paste("Marginals of X with p =", p, ", n =", n))
> curve(dnorm(x, 0, 1), add = TRUE, col = "red")
```

1a_hist_comp.png

Looking at the graph, it follows $N(0,1)$ pretty well.

(b)

```
> # 1. (b)
>
> library(mnormt)
>
> norm_sample = function(n, p, rho) {
+   # sigma as given in Exercise 7.5
+   I = diag(p)
+   J = matrix(1, p, p)
+   sigma = ((1 - rho) * I + rho * J)
+
+   rmnorm(n, mean = rep(0, p), varcov = sigma)
+ }
>
> system.time(gibbs_sample(n, p, rho))
   user  system elapsed
  0.003   0.000   0.002
> system.time(norm_sample(n, p, rho))
   user  system elapsed
  0.000   0.000   0.001
```

The rmnorm function seems to be much faster than the Gibbs sampling algorithm, althought this might not be as big of a difference at low iteration count.

2.

We aim to prove that

$$f(x,y) = \frac{f(y|x)}{\int \frac{f(y|x)}{f(x|y)} dy}$$

Let's start from the right-hand side and start evaluating from there:

$$
\begin{aligned}
&= \frac{f(y|x)}{\int \frac{f(y|x)}{f(x|y)} dy} && \text{Given} \\
&= \frac{f(y|x)}{\int \frac{f(y|x)}{\frac{f(x,y)}{f(y)}} dy} && \text{Bayes' theorem} \\
&= \frac{f(y|x)}{\int \frac{f(y|x) \cdot f(y)}{f(x,y)} dy} && \text{Arithmetic} \\
&= \frac{f(y|x)}{\int \frac{f(y|x) \cdot f(y)}{f(y|x) \cdot f(x)} dy} && \text{Definition of conditional probability} \\
&= \frac{f(y|x)}{\int \frac{f(y)}{f(x)} dy} && \text{Arithmetic} \\
&= \frac{f(y|x)}{\frac{1}{f(x)}} && \text{Integration} \\
&= f(y|x) \cdot f(x) && \text{Arithmetic} \\
&= f(x, y) && \text{Definition of conditional probability}
\end{aligned}
$$

Exactly as we sought to prove.